DYNAMIC PLANET

TEST

HAWK-HORNET INVITATIONAL

NOVEMBER 15, 2025

Team Name: ____

Team Number: _________

Rules:

- Each team may bring:
  - A binder of any size containing information in any form and from any source. Sheet protectors, lamination, tabs, and labels are permitted. No material may be removed from the binder throughout the event.
  - two Class II calculators (non-programmable, non-graphing calculators)

- You have 50 minutes to complete the test

- You may write on this test and pull it apart. **All answers must go on the answer sheet.**

Score: ______ / 93

Rank: ______

MULTIPLE CHOICE SECTION: Bubble in the best answer on your bubble sheet for each question. *(1 point each)*

1. The presence of a strong permanent thermocline in tropical oceans is a major factor that contributes to:
    a. Deep water formation and global circulation.
    b. High rates of primary productivity due to nutrient mixing.
    c. The stability of the water column and poor nutrient transfer to the surface.
    d. Massive seasonal temperature fluctuations at the surface.

2. Below the main thermocline, the deep ocean temperature is set by conditions at polar formation sites, but it is also subject to adiabatic heating as the water sinks and moves. This heating is a result of:
    a. Frictional energy dissipation from deep currents.
    b. Increased pressure compressing the water parcel.
    c. Absorption of residual infrared radiation from the surface.
    d. Geothermal heat flux from the seafloor.

3. What is the primary mechanism that causes the surface mixed layer to deepen and the seasonal thermocline to erode and disappear in temperate oceans during the winter months?
    a. Increased tidal current velocities.
    b. Increased longwave (IR) radiation from space.
    c. Net surface cooling and wind-driven turbulent mixing.
    d. Submarine volcanic heating.

4. Although river runoff contributes a vast amount of dissolved ions, the primary origin of the major dissolved ions (like Cl^- and Na^+) in the ocean is long-term:
    a. Volcanic outgassing and chemical weathering of the lithosphere.
    b. Biological respiration and excretion in the deep ocean.
    c. Sublimation of water vapor from the atmosphere into the ocean.
    d. The continuous dissolution of NaCl from continental shelf evaporite deposits.

5. In the water cycle, a high rate of evaporation at the ocean surface causes the local salinity to:
    a. Increase, because the salt is left behind as water vapor escapes.
    b. Decrease, because the vapor takes some salt with it.
    c. Change only in coastal areas, not in the open ocean.
    d. Remain unchanged, as evaporation only affects surface temperature.

6. Why is the density of the Deep Layer (80% of ocean volume) often considered relatively homogeneous?
    a. It is isolated from the extreme temperature and salinity changes occurring at the surface.
    b. Fast-moving currents stir the deep ocean violently.
    c. Biological activity constantly consumes dense plankton.
    d. The high pressure chemically alters the water to have a constant density.

7. Which process is primarily responsible for the low nutrient concentrations observed in the Surface Mixed Layer of the subtropical gyres?
    a. High rates of upwelling bringing deep water to the surface.
    b. The rapid input of nutrients from rivers.
    c. Biological consumption and stratification preventing deep nutrient resupply.
    d. Chemical precipitation of nutrients onto the seafloor.

8. What is the primary way that nutrients like nitrate (NO_3^-) and phosphate (PO_4^{3-}) are recycled from the surface back into the deep ocean?
    a. Through upwelling and turbulent mixing.
    b. Absorption by clay minerals and sinking to the seafloor.
    c. Through volcanic eruptions at mid-ocean ridges.
    d. The sinking of detritus and subsequent decomposition at depth.

9. During the day, phytoplankton photosynthesis in a coastal area consumes CO_2 dissolved in the water. This process causes the local water pH to:
   a. Increase.
   b. Decrease.
   c. Fluctuate rapidly between 7.0 and 8.5.
   d. Remain unchanged due to the buffering capacity of the ocean.

10. The massive spring bloom of phytoplankton in the North Atlantic occurs when which two conditions change simultaneously?
   a. Increased sunlight and the stabilization of the water column (stratification).
   b. Decreased salinity and extremely cold temperatures.
   c. Decreased wind speed and increased river runoff.
   d. Increased light intensity and increased nutrient input from upwelling.

11. Which of the following processes causes the movement of deepwater currents?
   a. prevailing winds
   b. the Coriolis Effect
   c. differences in temperature and salinity
   d. the shape of the ocean basin

12. The California Current is a cool, slow-moving current that flows along the western coast of North America. What is its primary effect on the climate of the adjacent land?
   a. It causes frequent tropical storms to form near the coast.
   b. It raises the average winter temperature, preventing snow and ice.
   c. It moderates the temperature, causing mild, often foggy conditions and low precipitation.
   d. It delivers high levels of moisture, causing heavy seasonal rainfall.

13. The strong, stable, and permanent thermocline found at the Equator is maintained primarily by the constant input of solar energy and:
   a. High surface salinity due to evaporation.
   b. Constant tidal energy dissipation.
   c. Deep-sea pressure effects which increase water density.
   d. The lack of strong Coriolis force to drive deep, large-scale current mixing.

14. Surface divergence in the ocean typically leads to which of the following processes?
   a. Downwelling and warmer surface temperatures.
   b. Increased concentration of floating debris.
   c. Upwelling and increased nutrient concentration at the surface.
   d. A depression of the sea surface height.

15. A surface convergence zone is often a region where floating materials (like plastic debris) accumulate. This is because
   a. The strong winds at a convergence zone blow the debris inward.
   b. Surface waters are flowing away from the central point.
   c. The horizontal water flow forces the materials to collect at the central point before being pushed down.
   d. The associated upwelling traps materials on the surface.

16. Which statement correctly describes the water temperature in a typical open-ocean surface divergence zone?
   a. The surface water is typically colder than the surrounding areas due to the upwelling of deep water.
   b. The temperature is highly variable but generally warmer than the coastal regions.
   c. The surface temperature is determined solely by the local air temperature, not the circulation.
   d. The surface water is typically warmer than the surrounding areas due to downwelling.

17. The Ekman layer depth is shallowest in which of these locations?
    a. In a calm, stratified coastal area near the equator.
    b. At the surface in the mid-latitude gyre center.
    c. In the open ocean at 60° North, during a summer storm.
    d. Near the pole during a persistent, strong wind event.

18. What is the primary consequence of Ekman convergence in the center of a subtropical gyre?
    a. Strong upwelling of nutrient-rich water.
    b. A mound of surface water (a sea surface 'hill') that drives geostrophic flow.
    c. Intense vertical mixing that breaks down the pycnocline.
    d. The cessation of all current movement due to balanced forces.

19. Which type of ocean current is primarily responsible for generating the largest, most energetic mesoscale eddies, often referred to as 'rings'?
    a. Western boundary currents.
    b. Equatorial counter-currents.
    c. Deep, thermohaline-driven currents.
    d. Eastern boundary currents.

20. Which of these oceanographic instruments is most commonly used to detect and track the movement of mesoscale eddies in the open ocean?
    a. Niskin bottles for discrete water sampling.
    b. Seismographs on the ocean floor.
    c. Deep-diving submersibles for visual observation.
    d. Satellite altimeters to measure sea surface height anomalies.

21. Which pair of currents are the primary mechanisms for transporting significant amounts of heat energy from the tropics toward the poles?
    a. The Deep Western Boundary Current and the Antarctic Circumpolar Current
    b. The California Current and the Humboldt Current
    c. The Gulf Stream and the Kuroshio Current
    d. Tidal currents and coastal upwelling

22. What is the defining characteristic that determines when a deep-water wave becomes a shallow-water wave?
    a. The wave period begins to decrease.
    b. The wave height reaches a maximum.
    c. The water depth is less than half the wave's wavelength.
    d. The wave's speed becomes equal to the wind speed.

23. As a wind wave approaches the shore and "feels bottom," its speed decreases, its wavelength decreases, and its height:
    a. Decreases
    b. Remains constant
    c. Increases
    d. Becomes equal to its period

24. The theoretical depth to which a deep-water wave causes water motion is called the wave base, and is equal to:
    a. The wave's height
    b. Twice the wave's period
    c. One-half the wave's wavelength
    d. The total depth of the ocean floor

25. How long does it take for a semidiurnal tide to go from high to low?
    a. 12 hours 25 min
    b. 3 hrs 14.5 min
    c. 24 hr 50 min
    d. 6 hrs 12.5 min

26. The side of the Earth *opposite* the Moon experiences a high tide because of:
    a. The constructive interference of solar tides.
    b. The centrifugal force of the Earth-Moon system.
    c. The strong direct gravitational pull of the Moon.
    d. Deep ocean wave reflection.

27. During a spring tide, the gravitational forces of the Moon and Sun are:
    a. Working against each other.
    b. Combining (in alignment) to create a maximum pull.
    c. Only pulling on the opposite side of Earth.
    d. Equal to the force of the Earth's rotation.

28. During a neap tide, the gravitational forces of the Moon and Sun are:
    a. Working at right angles to each other (in opposition).
    b. Combining to create a maximum pull.
    c. Completely canceled out, resulting in no tide.
    d. Stronger than the Moon's pull alone.

29. When comparing a spring high tide and a neap high tide, the spring high tide will be:
    a. Lower than the neap high tide.
    b. Higher than the neap high tide.
    c. Exactly the same height as the neap high tide.
    d. Longer in duration than the neap high tide.

30. The primary effect of tidal resonance in a confined basin is to:
    a. Eliminate the tides completely.
    b. Shift the timing of the tides by several hours.
    c. Amplify the tidal range (the difference between high and low tide).
    d. Change a semidiurnal tide into a diurnal tide.

31. An estuary is best defined as:
    a. A large, deep lake that connects to the ocean via a small river.
    b. A semi-enclosed body of water where fresh water from a river mixes with salt water from the ocean.
    c. A stretch of coast where only salt water is present due to high evaporation.
    d. A coastal lagoon separated from the ocean by a coral reef.

32. What is the primary environmental stressor for organisms living in an estuary?
    a. Consistent, high salinity
    b. Lack of dissolved oxygen
    c. Fluctuating salinity and temperature
    d. High light levels and UV exposure

33. The high productivity and rich fisheries found in upwelling zones (like the coast of Peru or California) are directly attributable to the supply of:
    a. Warm surface water.
    b. Nitrates and phosphates from deep water.
    c. Strong tidal currents.
    d. Low oxygen water.

34. A seamount whose top has been eroded flat by wave action and has since subsided (sunk) below the surface is called a:
    a. Sea stack   b. Abyssal hill   c. Mid-ocean ridge   d. Guyot

35. Sediment derived from the breakdown of rocks on continents and transported to the ocean by rivers, wind, or glaciers is classified as
    a. Hydrogenous sediment
    b. Biogenous sediment
    c. Cosmogenous sediment
    d. Lithogenous (Terrigenous) sediment

36. The Calcium Carbonate Compensation Depth (CCD) is the depth below which
    a. The rate of $CaCO_3$ dissolution exceeds the rate of $CaCO_3$ deposition.
    b. All deep-sea organisms die due to high pressure.
    c. Silica shells rapidly dissolve.
    d. The aphotic zone begins.

37. A major consequence of climate-driven sea level rise and increased storm intensity on coastal freshwater resources is:
    a. Increased oxygenation of groundwater.
    b. A decrease in groundwater temperature.
    c. A rise in the water table, filling coastal aquifers.
    d. Saltwater intrusion into coastal aquifers.

38. What is the collective term for the trace chemical or physical indicators preserved in sediment cores that are used to reconstruct past oceanographic or climatic conditions?
    a. Proxies
    b. Clastic materials
    c. Lithification products
    d. Hydrogenous nodules

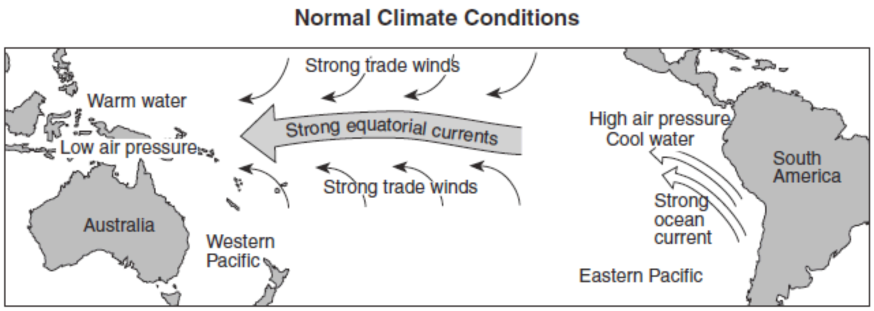
39. The calculated Salinity value derived from CTD measurements is reported in which unit?
    a. Parts per thousand (ppt)
    b. Siemens per meter (S/m)
    c. Practical Salinity Unit (PSU)
    d. Grams per kilogram (g/kg)

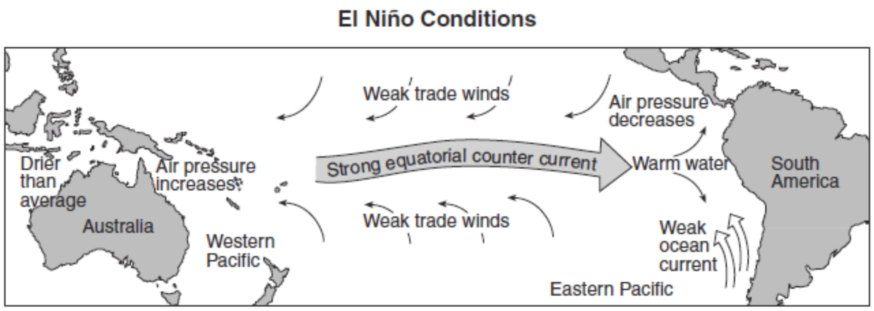
40. In a geostrophic current, if the isobars are closely spaced, the current velocity is typically:
    a. Zero
    b. Slow
    c. Fast
    d. Variable and unpredictable

41. As ocean waters warm, many mobile marine species (like fish and plankton) are observed migrating:
    a. Toward the equator and into deeper waters.
    b. Toward the poles and into deeper waters.
    c. Away from the poles and into shallower waters.
    d. Toward the equator and into shallower waters.

42. Oozes, which are deep-sea sediments composted of at least 30% skeletal remains of microscopic marine organism, are classified as
    a. Biogenous sediment.
    b. Volcanogenous sediment.
    c. Hydrogenous sediment.
    d. Lithogenous sediment.

Use the map below to answer the following 2 questions. The map shows normal and El Nino conditions of the Pacific Ocean.



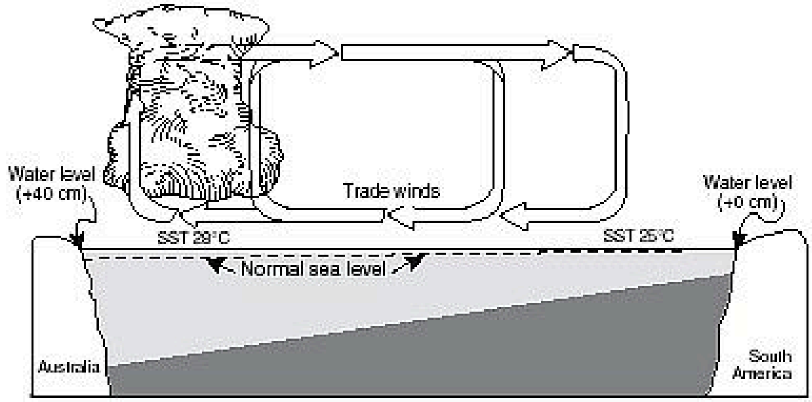


43. Under normal climate conditions, the characteristics of the surface ocean current that flows along most of the west coast of South America would be described as:
    a. cool water moving toward the equator
    b. cool water moving away from the equator
    c. warm water moving toward the equator
    d. warm water moving away from the equator

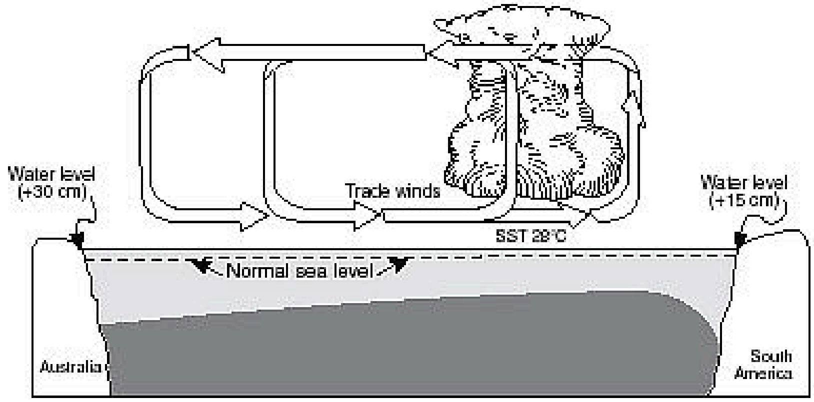
44. During El Nino conditions, air above the Pacific Ocean moving over the land on the equatorial west coast of South America is likely to be:
    a. cooler and drier than usual
    b. cooler and wetter than usual
    c. warmer and drier than usual
    d. warmer and wetter than usual

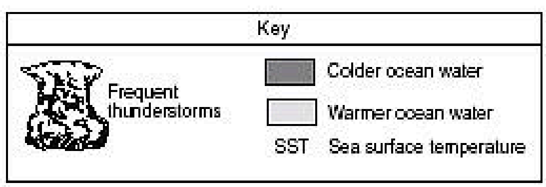
Base your answers to the next three questions on the two cross-sections below, which represent the Pacific Ocean and the atmosphere near the Equator during normal weather (cross section A) and during El Nino conditions (cross section B). Sea surface temperatures (SST) are labeled and trade wind directions are shown with arrows. Cloud buildup indicates regions of frequent thunderstorm activity. The change from normal sea level is shown at the side of each diagram.

Cross Section A: Normal Weather



Cross Section B: El Niño Conditions





45. Which statement correctly describes sea surface temperatures along the South American coast and Pacific trade winds during El Nino conditions?
    a. the sea surface temperatures are warmer than normal and Pacific trade winds are from the west
    b. the sea surface temperatures are warmer than normal and Pacific trade winds are from the east
    c. the sea surface temperatures are cooler than normal and Pacific trade winds are from the west
    d. the sea surface temperatures are cooler than normal and Pacific trade winds are from the east

46. Compared to normal weather conditions, the shift of the trade winds caused sea levels during El Nino conditions to:
    a. decrease at both Australia and South America
    b. decrease at Australia and increase at South America
    c. increase at Australia and decrease at South America
    d. increase at both Australia and South America

47. The development of El Nino conditions over this region of the Pacific Ocean has caused:

a. changes in worldwide precipitation patterns
b. the reversal of Earth's seasons
c. increased worldwide volcanic activity
d. decreased ozone levels in the atmosphere

48. La Niña is caused by
    a. A build up of warmer than normal waters in the tropical Pacific
    b. A build up of cooler than normal waters in the tropical Atlantic
    c. A build up of cooler than normal waters in the tropical Pacific
    d. A build up of cooler than normal waters in the tropical Atlantic

49. Sediment composed of tiny metallic spherules and silicate dust that enters the ocean from outer space is classified as:
    a. Biogenous sediment c. Hydrogenous sediment
    b. Cosmogenous sediment d. Evaporite sediment

50. What type of magma is the seafloor formed from?
    a. Rhyolitic b. Basaltic c. Andesitic d. None of the above

51. Coastal upwelling, driven by offshore Ekman transport, impacts the local temperature profile by causing the thermocline to:
    a. Shoal (move closer to the surface).
    b. Disappear entirely due to constant mixing.
    c. Deepen and become weaker (less gradient).
    d. Invert, with cold water on top of warm water.

52. Which of the following best describes the direction of a longshore current?
    a. Parallel to the shoreline.
    b. Perpendicular to the shoreline, flowing offshore.
    c. Perpendicular to the shoreline, flowing onshore.
    d. Rotational (circular) offshore.

53. A large, continuous coral reef structure separated from the mainland or island by a deep lagoon is known as a:
    a. Fringing reef b. Barrier reef c. Guyot d. Atoll

54. The period when the tidal current ceases (stops moving) as the tide changes direction is called:
    a. Tidal bore b. High current c. Slack water d. Mean tidal level

TRUE/FALSE SECTION: For each of the following statements, bubble in (T) if it is TRUE and (F) if it is FALSE *(1 point each)*

55. The deep ocean can reach temperatures under zero degrees celsius and remain in a liquid state.
56. Coastal protection structures such as groynes and jetties are built to accelerate (increase) coastal erosion.
57. The Intermediate Layer of the three-layer model is where the pycnocline is typically located.
58. Objects are deflected to the left in the southern hemisphere due to the Coriolis effect.
59. The continental shelf is part of the oceanic crust.
60. True or False: Because of Ekman transport, persistent winds blowing parallel to a coastline can cause convergence along the coast.
61. Generally the fastest and deepest ocean currents are the western boundaries.
62. Coastal upwelling fronts form when cold, deep water rises to the surface, creating a temperature contrast with warmer offshore surface water.
63. A wave with a wavelength of 200 meters traveling in water 101 meters deep is classified as a deep-water wave.
64. The continental slope is generally wider and less steep along an active continental margin than a passive continental margin.
65. Primary productivity is high in the neritic zone.
66. Groynes successfully stop beach erosion along the entire stretch of coastline where they are built.
67. The mid-ocean ridge is entirely volcanic and completely devoid of any form of life.
68. Ocean acidification primarily affects calcifying organisms and has no significant documented impact on non-calcifying organisms like fish.
69. The speed of sound in seawater is primarily a function of water temperature, salinity, and pressure.
70. Equatorial Kelvin waves play a crucial role in the dynamics of the El Niño-Southern Oscillation (ENSO).
71. Rossby waves are a type of shallow water wave whose speed is independent of the depth of the ocean.
72. The continental shelf is relatively narrow along the active Pacific coast of North and South America.
73. Rip currents are usually strongest when waves are small and the tide is low.
74. A marine terrace is a depositional feature composed of fine-grained mud from an estuary.
75. In an area with a mixed semidiurnal tidal pattern, the tidal currents will have four distinct periods of maximum velocity and four distinct periods of slack water during a lunar day.
76. Estuaries are typically located in regions with high wave energy, making them ideal for coastal fishing.

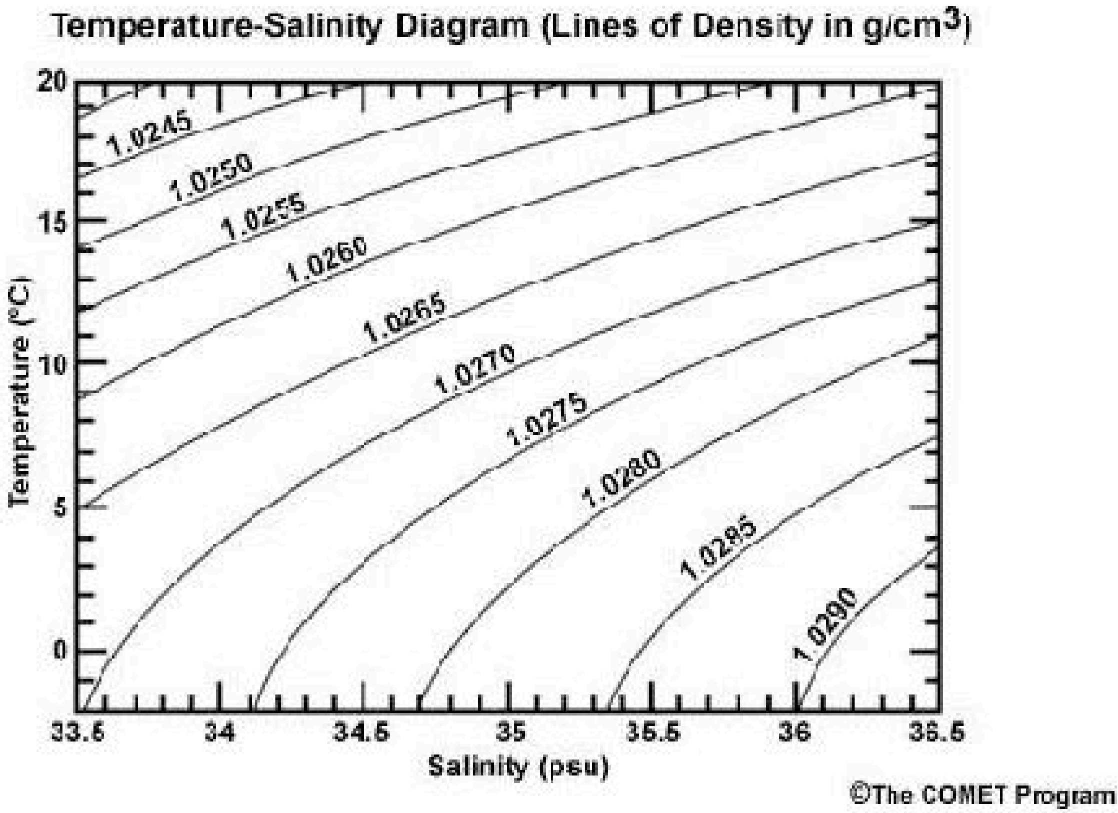
SHORT ANSWER SECTION: Answer each of the questions below in the appropriate spaces on your answer sheet.

77. Draw a temperature vs depth graph. Label the 3 parts. *(4 points)*

78. Use the following four chemical reactions to both explain how increasing levels of CO_2 relates to ocean acidification and identify how an overabundance of CO_2 leads to the stymieing of coral growth. *(4 points)*

$H^+ + CO_3^{2-} \rightarrow HCO_3^-$	$H_2CO_3 \rightarrow H^+ + HCO_3^-$	$CO_3^{2-} + Ca^{2+} \rightarrow CaCO_3$	$CO_2 + H_2O \rightarrow H_2CO_3^-$

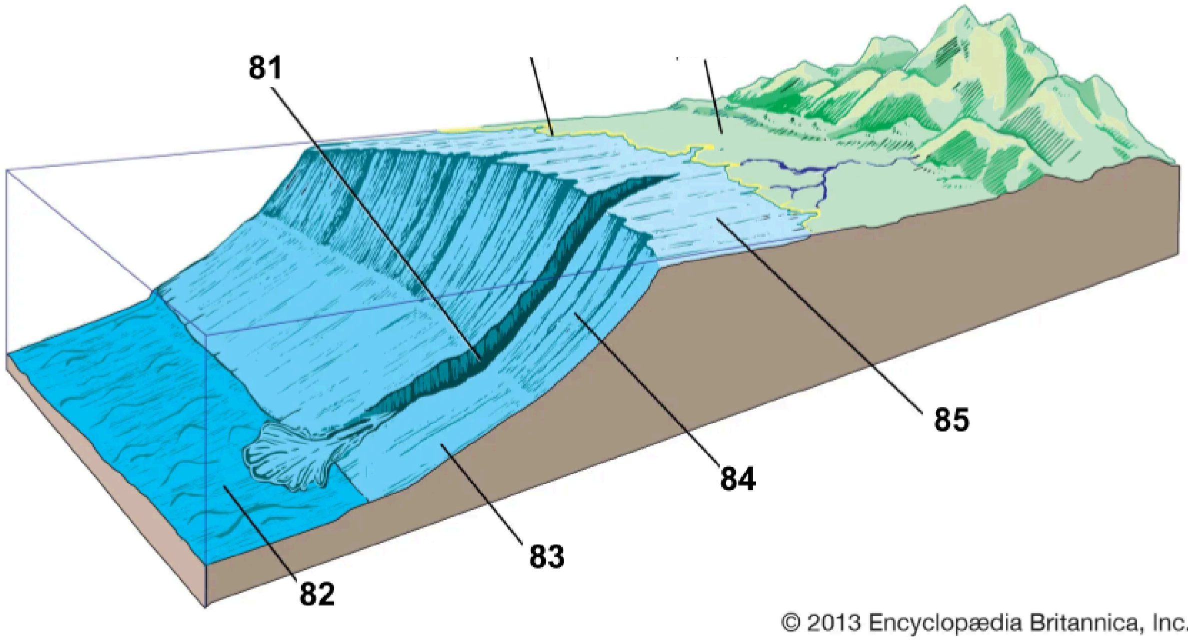
Use the diagram below to answer the next two questions: *(2 points each)*



79. The temperature is 15℃ and the salinity is 35psu, so the density is ____.

80. The temperature is ___ and the salinity is 36.1psu, so the density is 1.0290g/cm^3.

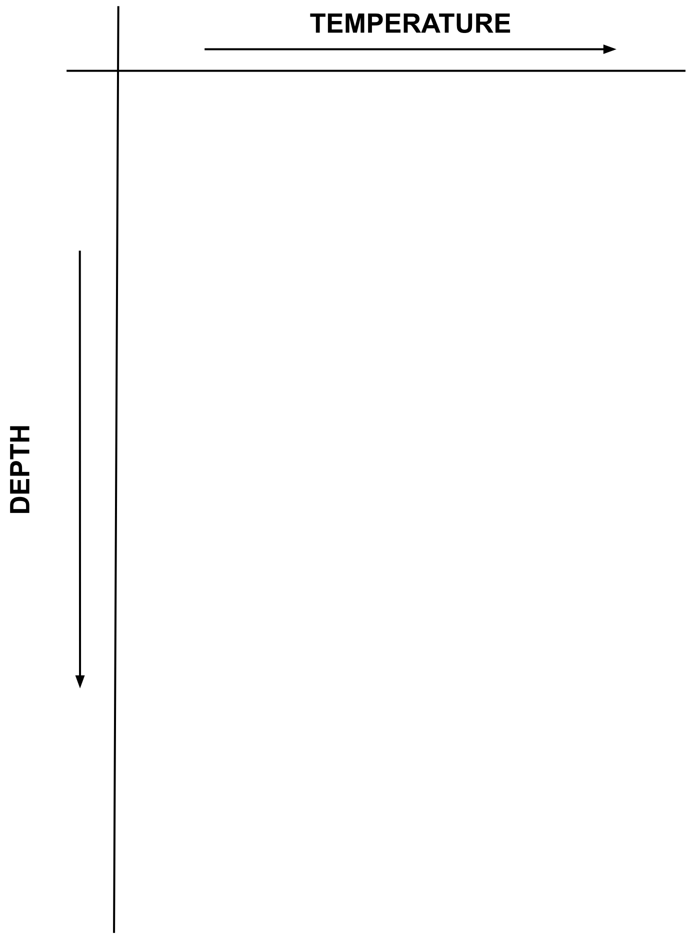
#81-85: Label the diagram



Team Name: ____ Team Number: ______________

SHORT ANSWER SHEET

77.



78. ____

OVER →

79. ______________________________

80. ______________________________

81. ______________________________

82. ______________________________

83. ______________________________

84. ______________________________

85. ______________________________